Forests and Deforestation

Introduction to Forests

A forest is a large area dominated by trees, together with the plants, animals, fungi and microorganisms that live among them. Forests cover about 31 percent of the Earth's land area, about 4.1 billion hectares, according to the Food and Agriculture Organization (FAO). More than half of the world's forests are in just five countries: Russia, Brazil, Canada, the United States and China. The main types are tropical rainforests, such as the Amazon and the Western Ghats, temperate forests, boreal forests or taiga, mangroves, and dry deciduous forests. Forests are home to most of the world's land-based plant and animal species, and they provide timber, food, medicine, fuel and livelihoods for hundreds of millions of people.

Why Are Forests Called the Lungs of the Earth?

Forests are called the lungs of the Earth because, like lungs, they help the planet "breathe". Through photosynthesis, trees absorb carbon dioxide from the air and, using sunlight, convert it into sugars and release oxygen. In this way forests clean the air, store large amounts of carbon in their wood, leaves and soil, and help to regulate the climate. The Amazon rainforest is often called the lungs of the planet. However, scientists point out an important nuance: forests also use oxygen for their own respiration and the decay of dead matter, so the net oxygen they add to the atmosphere is small, and about half of the world's oxygen production comes from phytoplankton in the oceans. The real importance of forests is that they act as carbon sinks, locking away carbon dioxide that would otherwise warm the planet, and when they are cut or burned, this carbon is released.

Other Benefits of Forests

Forests regulate the water cycle by releasing moisture through transpiration, which helps produce rain, and by holding rainwater in the soil and feeding springs and rivers. Tree roots prevent soil erosion, landslides and floods. Forests support biodiversity, provide shade and cooling, reduce noise and dust, supply forest products such as honey, bamboo, lac and medicinal plants, and have spiritual and cultural importance for many tribal communities.

What Is Deforestation?

Deforestation is the permanent removal of forests so that the land can be used for other purposes. The FAO estimates that the world lost about 420 million hectares of forest between 1990 and 2020, and about 10 million hectares are cleared every year, an area about the size of Kerala and Tamil Nadu together. Tropical forests in Brazil, Indonesia and the Democratic Republic of the Congo suffer most.

What Is the Main Cause of Deforestation?

The main cause of deforestation is agricultural expansion. The FAO and other studies estimate that about 90 percent of deforestation is linked to clearing land for farming, in particular for cattle ranching, soybean cultivation, oil palm plantations and other crops. Other causes include commercial logging and illegal timber trade, collection of firewood, mining, building of roads, dams and cities, and forest fires, which are made worse by drought and climate change. In India, causes include shifting cultivation in the north-east, encroachment, infrastructure projects, mining and forest fires.

Effects of Deforestation

The loss of forests releases stored carbon and contributes to climate change, destroys habitats and causes extinction of species, increases soil erosion, landslides and flooding, disturbs rainfall patterns, and harms indigenous peoples whose lives depend on forests. Deforestation also increases the risk of diseases passing from animals to humans, since wildlife and people come into closer contact.

What Are Afforestation and Reforestation?

Afforestation is the planting of trees on land that was not recently forest, such as barren land, wasteland or degraded grassland, to create a new forest. Reforestation is the replanting of trees in an area where forest was cut down or destroyed, to restore it. Both increase carbon absorption and restore ecosystems. Related ideas are agroforestry, which combines trees with crops, and natural regeneration, in which forests are allowed to regrow by protecting the land. The Miyawaki method, developed by Japanese botanist Akira Miyawaki, plants dense native mini-forests, and has been used in Chennai and many Indian cities.

Forests in India

According to the India State of Forest Report 2023, released in December 2024 by the Forest Survey of India, forest and tree cover together make up about 25.17 percent of India's geographical area, with forest cover alone at about 21.76 percent and tree cover about 3.41 percent. India's National Forest Policy of 1988 aims to bring one-third of the country's area under forest or tree cover. The Chipko Movement, which began in 1973 in Uttarakhand, with villagers, especially women, hugging trees to stop logging, became famous worldwide, and Sunderlal Bahuguna and Gaura Devi are linked with it. Van Mahotsav, the tree-planting festival, was started in 1950 by K. M. Munshi, and the Forest Conservation Act was passed in 1980. The government launched the campaign "Ek Ped Maa Ke Naam" (a tree in the name of mother) in June 2024. The Western Ghats and the Nilgiris in Tamil Nadu hold some of the richest forests. World Forest Day is on 21 March.

Challenges and Future

Solutions include protecting existing forests, sustainable forestry certification, restoring degraded land, community forest management, reducing demand for products linked to forest loss, and technology such as satellite monitoring. The Glasgow Declaration of 2021 pledged to end deforestation by 2030.

Quick Facts

- Forests cover about 31 percent of the world's land and act as carbon sinks, which is why they are called the lungs of the Earth.
- The main cause of deforestation is clearing land for agriculture, about 90 percent of it.
- Afforestation is planting trees on land that was not forest; reforestation is replanting where forests were lost.
- India's forest and tree cover is about 25.17 percent of its area (ISFR 2023).
- The Chipko Movement began in 1973 in Uttarakhand.